SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1005	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	Akhila Pochimireddy	**Reg. No.:**	192425225
Date:	13-10-2026	**Duration:**	60 minutes

Code of Conduct

I **Akhila Pochimireddy(192425225)***certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Akhila.P

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. Identify the Project Vision and Stakeholders

Project Vision

To develop a user-friendly and transparent online agricultural marketplace that connects farmers directly with wholesalers, retailers, and consumers. The platform will allow farmers to list available produce, buyers to discover products and place orders, and all participants to follow order status, delivery progress, payments, and transaction records.

Objectives

- Register and authenticate farmers and buyers.
- Allow farmers to create, update, and remove agricultural product listings.
- Show product availability, quantity, price, and basic product details.
- Enable buyers to place orders and view order status.
- Support secure payment processing through a payment service.
- Track delivery status from dispatch to delivery.
- Maintain transaction history and provide market analytics.
- Collect ratings and feedback to improve trust and service quality.

Stakeholders and Their Roles

Stakeholder	Role	Main Interest
Farmers	Register, manage produce listings, confirm orders, prepare goods	Fair visibility, wider market access, clear prices and payments
Wholesalers	Buy produce in bulk and monitor orders/deliveries	Availability, quantity, reliable fulfilment
Retailers	Purchase produce for resale	Consistent supply and transparent pricing
Consumers	Browse products, order produce, pay and rate purchases	Convenience, product details, delivery updates
Delivery partners	Collect, transport and update delivery status	Accurate delivery details and efficient routing
Platform administrator	Manage accounts, moderate listings, resolve issues and view reports	Security, reliable operation and policy compliance
Payment service provider	Process payment requests and return payment status	Secure, valid transactions
Development team / product owner	Build, test, prioritize and maintain the platform	Deliver useful features within sprint goals

2. Create Epics and User Stories

Epic	ID	User Story	Priority	Points
E1 – Account Management	US1	As a farmer, I want to register an account, so that I can offer my produce on the platform.	High	3
E1 – Account Management	US2	As a buyer, I want to register and sign in, so that I can place and manage orders.	High	3
E2 – Product Listings	US3	As a farmer, I want to add a product with name, quantity, price, and availability, so that buyers can discover my produce.	High	5
E2 – Product Listings	US4	As a buyer, I want to browse and filter available products, so that I can find produce matching my needs.	High	5

E3 – Order Management	US5	As a buyer, I want to place an order for an available product, so that I can purchase produce conveniently.	High	5
E3 – Order Management	US6	As a farmer, I want to view and update order status, so that I can manage fulfilment.	High	3
E4 – Payments	US7	As a buyer, I want to pay through a secure payment option, so that my transaction can be completed safely.	Medium	8
E5 – Delivery Tracking	US8	As a buyer, I want to view delivery status, so that I know when my order is expected to arrive.	Medium	5
E6 – Transaction History & Analytics	US9	As a user, I want to view previous transactions, so that I can review my purchases or sales.	Medium	5
E7 – Ratings & Feedback	US10	As a buyer, I want to rate a completed order, so that I can share feedback and help others make decisions.	Low	3
E6 – Transaction History & Analytics	US11	As an administrator, I want to view sales and demand summaries, so that I can understand platform activity.	Low	5

3. Define Acceptance Criteria

US1 – Farmer registration

Given a farmer is on the registration page, when valid required details and a unique contact/email are submitted, then an account is created and a success message is shown.

Given required fields are missing or the contact/email is already registered, when the form is submitted, then the system displays a clear validation message and does not create a duplicate account.

US2 – Buyer registration and sign-in
Given a buyer enters valid registration details, when the form is submitted, then the buyer account is created.
Given a registered buyer enters correct credentials, when they select Sign In, then the system opens their account; incorrect credentials show an error without granting access.
US3 – Add product listing
Given a signed-in farmer opens Add Product, when valid product name, quantity, unit, price, and availability are submitted, then the listing is saved and displayed.
Given quantity or price is zero/negative or a required field is blank, when the farmer submits the listing, then validation prevents saving and identifies the invalid field.
US4 – Browse products
Given active listings exist, when a buyer opens the marketplace, then each listing displays product name, price, available quantity, and farmer/seller details.
Given the buyer selects a product filter or search term, when the search is applied, then only matching available listings are shown.
US5 – Place order
Given a buyer selects an available listing and a valid quantity, when the buyer confirms the order, then an order record is created with a unique order ID and total amount.
Given requested quantity exceeds available stock, when the buyer tries to confirm, then the order is blocked and the available quantity is shown.
US6 – Update order status
Given a farmer has received an order, when the farmer marks it accepted, packed, or dispatched, then the status and update time are saved and visible to the buyer.
Given a user is not authorized for the order, when they attempt to change its status, then the system rejects the change.
US7 – Secure payment
Given an order is ready for payment, when the buyer completes payment through the payment provider, then the platform records the provider's success/failure result against the order.
Given payment fails or is cancelled, when the provider returns the result, then the order is not marked as paid and the buyer sees the payment status.
US8 – Delivery tracking
Given an order has been dispatched, when the delivery partner updates its status, then the buyer can view the latest status and update time.
Given no dispatch update exists, when the buyer opens tracking, then the system displays the current order stage rather than inventing a delivery location.
US9 – Transaction history
Given a signed-in user has completed or attempted transactions, when they open transaction history, then only their own transactions are displayed with date, order ID, amount, and status.
US10 – Rating and feedback
Given an order is marked delivered, when the buyer submits a rating within the allowed range and optional feedback, then the feedback is saved against that order.
Given an order is not delivered or the rating is outside the allowed range, when feedback is submitted, then the system rejects it with a helpful message.
US11 – Market analytics
Given transaction and listing data exists for a selected period, when an authorized administrator opens analytics, then the dashboard shows totals such as orders, sales value, and popular products for that period.

4. Prioritize the Product Backlog

Prioritization uses MoSCoW-style business importance and dependencies. Registration and product discovery are placed first because ordering depends on trusted accounts and available listings. Payments, delivery, analytics, and feedback follow once the basic order flow works.

Rank	Story	Priority	Points	Reason / Dependency
1	US1 Farmer registration	Must	3	Needed before farmers can create listings
2	US2 Buyer registration/sign-in	Must	3	Needed to place and manage orders
3	US3 Add product listing	Must	5	Core marketplace function; depends on farmer account
4	US4 Browse/filter products	Must	5	Buyers need to discover available produce
5	US5 Place order	Must	5	Core transaction flow; depends on accounts and listings
6	US6 Update order status	Should	3	Supports fulfilment and buyer visibility
7	US7 Secure payment	Should	8	Important for transaction completion; depends on order flow and provider integration
8	US8 Delivery tracking	Should	5	Depends on order dispatch and delivery updates
9	US9 Transaction history	Should	5	Requires saved order/transaction records
10	US10 Ratings and feedback	Could	3	Most useful after successful delivery
11	US11 Market analytics	Could	5	Depends on sufficient listing and transaction data

5. Plan the Sprint Backlog

Sprint Details

Sprint duration: 2 weeks (10 working days). Team capacity assumption: a small student team with approximately 20–25 story points of capacity. This is a planning assumption and should be adjusted to the actual team size and availability.

Sprint 1 selection: US1 Farmer registration (3 points), US2 Buyer registration/sign-in (3 points), and US3 Add product listing (5 points). Total: 11 story points. US3 depends on a working farmer account, so account work is planned first.

Actionable Development Tasks

US1 – Farmer registration (3 points)

Confirm required farmer fields and validation rules.
Design registration form and database fields.
Implement farmer registration and duplicate-contact validation.
Test valid, missing-field, and duplicate-account cases.

US2 – Buyer registration/sign-in (3 points)

Define buyer registration and sign-in screens.
Implement account creation and credential validation.
Apply role-based access for buyer and farmer accounts.
Test successful sign-in, invalid credentials, and access restrictions.

US3 – Add product listing (5 points)

Define product data fields: name, category, quantity/unit, price, description, availability.
Design Add Product form and product listing view.
Implement saving and displaying a farmer's product listing.
Validate required fields, quantity, and price.
Test that only an authenticated farmer can add or edit their own listings.

Sprint-wide tasks

Create a shared repository and agree coding conventions.
Conduct daily stand-ups and update the task board.
Perform peer review and integration testing.
Demonstrate completed stories during the sprint review and record feedback.

6. Estimate Effort Using Story Points

The Fibonacci sequence (1, 2, 3, 5, 8, 13) is used to estimate relative effort, complexity, and uncertainty. Estimates are illustrative Planning Poker values for a student project, not fixed hours.

Story	Points	Relative effort	Justification
US1 Farmer registration	3	Small	A straightforward form, field validation, and account persistence.
US2 Buyer registration/sign-in	3	Small	Similar form work plus credential checks and role access.
US3 Add product listing	5	Medium	Several product fields, validation, database persistence, and listing display.
US4 Browse/filter	5	Medium	Listing display,

products			search/filter logic, and availability handling.
US5 Place order	5	Medium	Quantity checks, order creation, total calculation, and order ID.
US6 Update order status	3	Small	A limited set of status transitions and authorization checks.
US7 Secure payment	8	Large	External payment integration, callback handling, failure cases, and security considerations.
US8 Delivery tracking	5	Medium	Status updates, timestamps, and buyer-facing tracking view.
US9 Transaction history	5	Medium	Filtering and displaying user-specific transaction records.
US10 Ratings and feedback	3	Small	Rating validation, feedback storage, and order eligibility checks.
US11 Market analytics	5	Medium	Aggregating listing and transaction data and displaying summaries.

7. Present the Sprint Goal and Expected Deliverables

Sprint Goal

Enable farmers and buyers to create accounts and allow an authenticated farmer to publish a validated agricultural product listing, establishing the foundation for the marketplace.

Expected Deliverables at the End of Sprint 1

A farmer registration screen with required-field and duplicate-account validation.
A buyer registration and sign-in flow with basic access control.
An Add Product screen for authenticated farmers.
A product record structure storing product name, quantity, unit, price, and availability.
A basic view that displays saved product listings.
Test cases and test results for registration, sign-in, and product listing.
A short sprint review demonstration and an updated product backlog based on feedback.

Definition of Done

The feature meets its acceptance criteria.
Input validation and relevant error messages are implemented.
Code is reviewed and integrated with the main project.
Relevant functional tests pass.
No known critical issue blocks the demonstration.
The team can demonstrate the feature during sprint review.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25